

ACCT102 Management Accounting

Week 5

Absorption costing and Variable costing

Learning Objective

Explain how variable costing differs from absorption costing and compute unit product costs under each method.

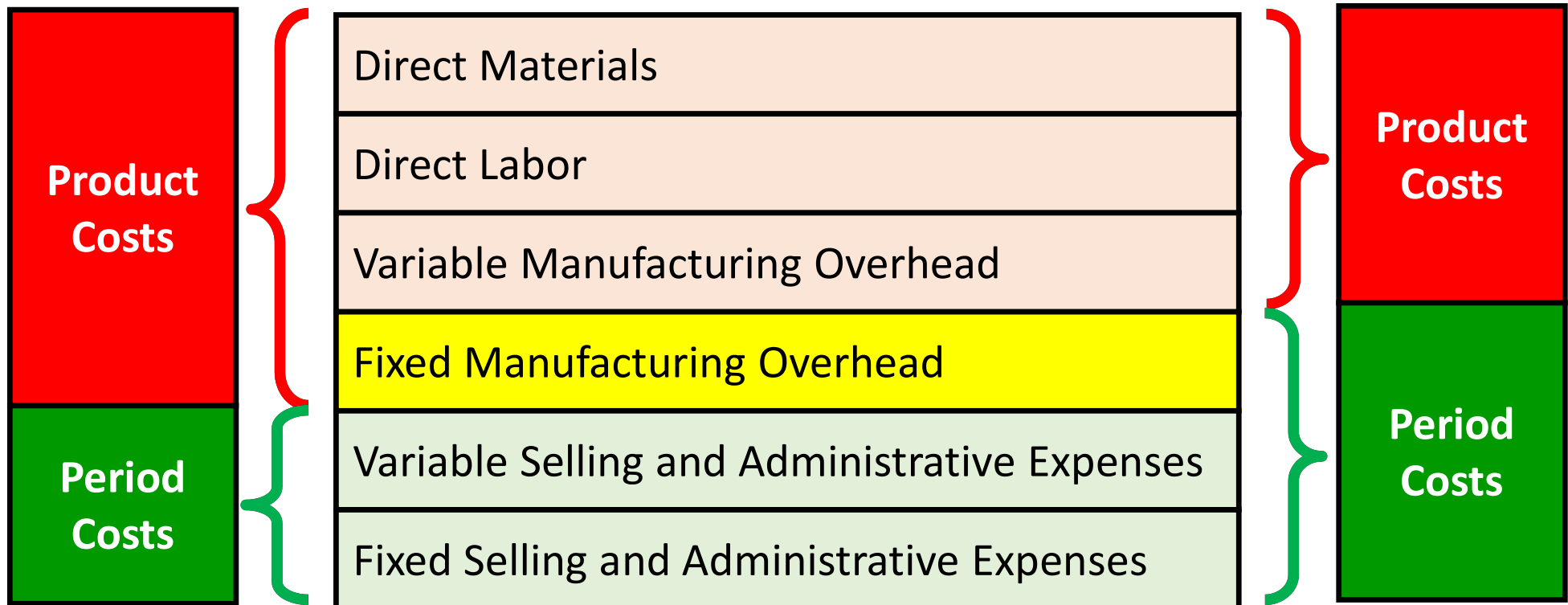
Key assumption: The following slides use actual costing, instead of normal costing.

Variable and Absorption Costing

- Under absorption costing, all production costs, variable and fixed, are included when determining unit product cost.
- Under variable costing, only the variable production costs are included in product costs.

Absorption Costing

Variable Costing



Variable versus Absorption Costing

Absorption Costing

- Matching principle - Fixed manufacturing costs must be assigned to products to properly match revenues and costs.

Variable Costing

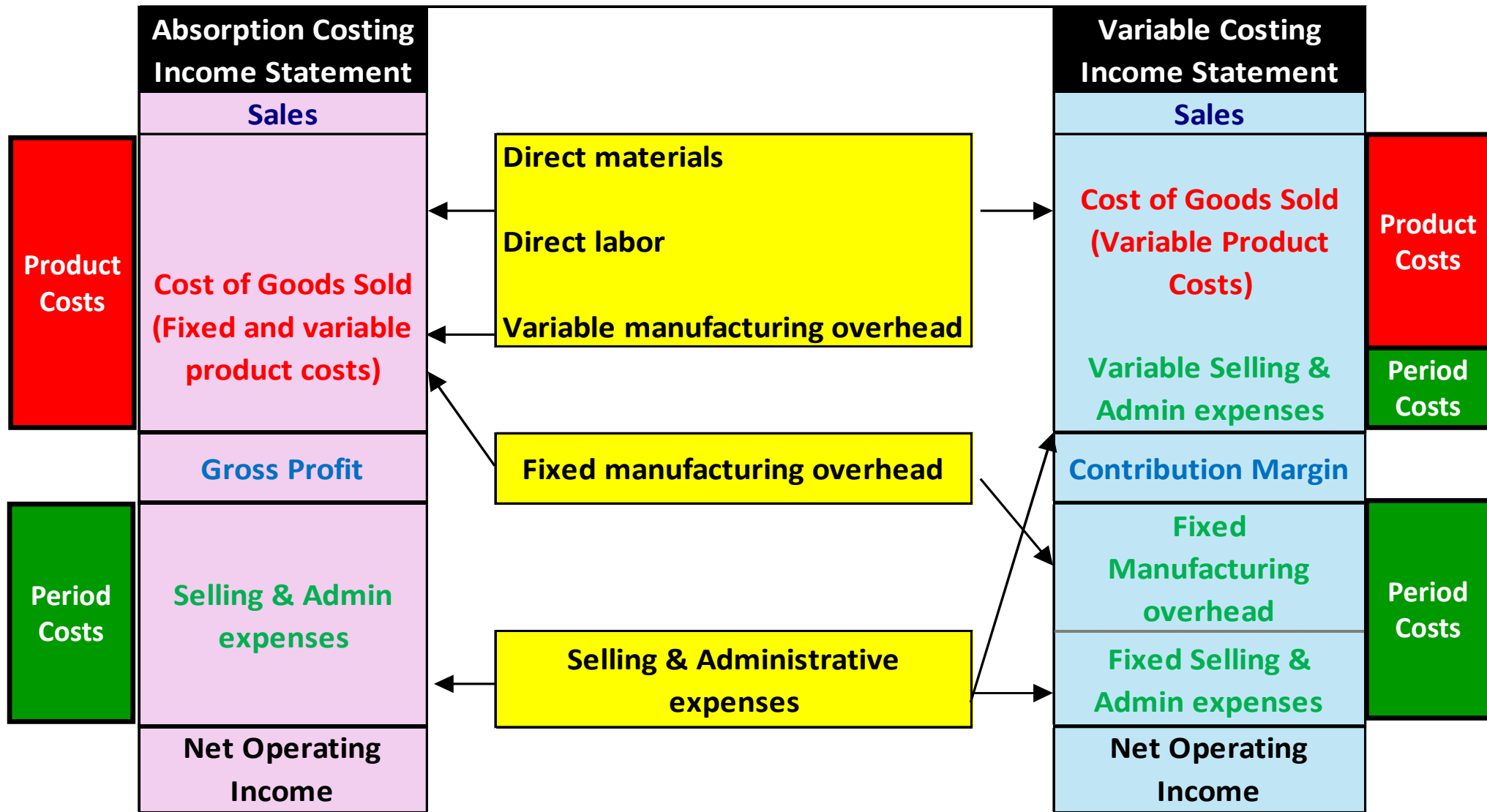
- Fixed manufacturing costs are capacity costs and will be incurred even if nothing is produced.

Variable and Absorption Costing Income Statement

Absorption Costing Income Statement
Sales
Cost of Goods Sold
Gross Profit
Selling & Admin expenses
Net Operating Income

Variable Costing Income Statement
Sales
Cost of Goods Sold
Variable Selling & Admin expenses
Contribution Margin
Fixed Manufacturing overhead
Fixed Selling & Admin expenses
Net Operating Income

Variable and Absorption Costing Income Statement



Quick Check 5A.1

Which method will produce the highest values for work in process and finished goods inventories?

- a. Absorption costing.
- b. Variable costing.
- c. They produce the same values for these inventories.
- d. It depends...

Unit Product Cost

Harvey Company produces 25,000 units of a single product annually.

Number of units produced annually	25,000
Variable costs per unit:	
Direct materials, direct labor, and variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3
Fixed costs per year:	
Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

Unit Product Cost Computations

Harvey Company produces 25,000 units of a single product annually.

Number of units produced annually	25,000
Variable costs per unit:	
Direct materials, direct labor, and variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3
Fixed costs per year:	
Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

<u>Unit product cost</u>	<u>Absorption Costing</u>	<u>Variable Costing</u>
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead	6	-
Unit product cost	\$ 16	\$ 10

Learning Objective

Prepare income statements using both variable and absorption costing.

Reconcile variable costing and absorption costing net operating incomes and explain why the two amounts differ.

Variable and Absorption Costing Income Statements (Y1)

- In Year 1, Only 20,000 units were sold during the year, at a price of \$30 each.
- There is no beginning inventory.

Number of units produced annually	25,000
Variable costs per unit:	
Direct materials, direct labor, and variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3
Fixed costs per year:	
Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

<u>Unit product cost</u>	<u>Absorption Costing</u>	<u>Variable Costing</u>
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead (\$3 x 2 DLH per unit)	6	-
Unit product cost	\$ 16	\$ 10

Variable Costing Income Statement (Y1)

	Variable Costing	
Sales (20,000 × \$30)		\$ 600,000
Less variable expenses:		
Variable cost of goods sold (20,000 × \$10)	\$ 200,000	
Variable selling & administrative expenses (20,000 × \$3)	60,000	
Total variable expenses		260,000
Contribution margin		340,000
Less fixed expenses:		
Fixed manufacturing overhead	\$ 150,000	
Fixed selling & administrative expenses	100,000	250,000
Net operating income		\$ 90,000

← Variable manufacturing costs only.

← Expense all fixed manu OH

Absorption Costing Income Statement (Y1)

Absorption Costing			
Sales (20,000 × \$30)		\$	600,000
Cost of goods sold (20,000 × \$16)			320,000
Gross margin			280,000
Less selling & administrative expenses:			
Variable (20,000 × \$3)	\$	60,000	
Fixed		100,000	160,000
Net operating income		\$	120,000

← Includes fixed manu OH

The rest of the fixed manufacturing overhead is deferred in inventory.
 5,000 units (produced but not sold) × \$6 = \$30,000.

Comparing the Two Methods (Y1)

- Under VC, all fixed manufacturing overheads are expensed during the period.
- Under AC, fixed manufacturing overheads are allocated to product costs.
 - Some of the fixed manufacturing overhead is deferred in Inventory, as some products are not sold (and in Inventory ending balance),

	Cost of Goods Sold	Ending Inventory	Period Expense	Total
Absorption costing				
Variable mfg. expenses	\$ 200,000	\$ 50,000	\$ -	\$ 250,000
Fixed mfg. expenses	120,000	30,000	-	150,000
	<u>\$ 320,000</u>	<u>\$ 80,000</u>	<u>\$ -</u>	<u>\$ 400,000</u>
Variable costing				
Variable mfg. expenses	\$ 200,000	\$ 50,000	\$ -	\$ 250,000
Fixed mfg. expenses	-	-	150,000	150,000
	<u>\$ 200,000</u>	<u>\$ 50,000</u>	<u>\$ 150,000</u>	<u>\$ 400,000</u>

Reconciling the Two Methods (Y1)

Variable costing net operating income	\$ 90,000
Add: Fixed mfg. overhead costs deferred in inventory	
(5,000 units × \$6 per unit)	30,000
Absorption costing net operating income	\$ 120,000

Produced but not sold × Fixed OH per unit

Variable and Absorption Costing Income Statements (Y2)

- In Year 2, 30,000 units were sold during the year, at a price of \$30 each.
- There is a beginning inventory of 5,000 units from Year 1.

	YEAR 1	YEAR 2
Number of units produced	25,000	25,000
Number of units sold	20,000	30,000
Units in beginning inventory	-	5,000
Unit sales price	\$ 30	\$ 30
Variable costs per unit:		
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Selling & administrative expenses	\$ 3	\$ 3
Fixed costs per year:		
Manufacturing overhead	\$ 150,000	\$ 150,000
Selling & administrative expenses	\$ 100,000	\$ 100,000

	AC	VC
Unit product cost	\$16	\$10

Variable Costing Income Statement (Y2)

	Variable Costing		
Sales (30,000 × \$30)		\$ 900,000	
Less variable expenses:			
Variable cost of goods sold (30,000 × \$10)	\$ 300,000		← <i>Variable manufacturing costs only.</i>
Variable selling & administrative expenses (30,000 × \$3)	90,000		
Total variable expenses		390,000	
Contribution margin		510,000	
Less fixed expenses:			
Fixed manufacturing overhead	\$ 150,000		← <i>Expense all fixed manu OH</i>
Fixed selling & administrative expenses	100,000	250,000	
Net operating income		260,000	

Absorption Costing Income Statement (Y2)

Absorption Costing			
Sales (30,000 × \$30)		\$	900,000
Cost of goods sold (30,000 × \$16)	← <i>Includes fixed manu OH</i>		480,000
Gross margin			420,000
Less selling & administrative expenses:			
Variable (30,000 × \$3)		\$	90,000
Fixed		100,000	190,000
Net operating income		\$	230,000

Fixed manufacturing overhead is released from beginning inventory balance 5,000 units (sold in Y2 but produced in Y1) × \$6 = \$30,000.

Reconciling the Two Methods (Y2)

Variable costing net operating income	\$ 260,000
Deduct: Fixed manufacturing overhead costs released from inventory	
→ (5,000 units × \$6 per unit)	30,000
Absorption costing net operating income	<u>\$ 230,000</u>

Previously produced but now sold x Fixed OH per unit

Reconciling the Two Methods (Summary)

No beginning inventory at Y1

No ending inventory at Y2

Costing Method	1st Period	2nd Period	Total
Absorption	\$ 120,000	\$ 230,000	\$ 350,000
Variable	90,000	260,000	350,000

Production vs Sales	Effect on inventory	VC profit vs AC profit
Produced = Sold	No change in inventory	AC = VC
Produced > Sold	Inventory increases <i>Some FMOH deferred in inventory for AC</i>	AC > VC
Produced < Sold	Inventory decreases <i>Some FMOH released from inventory for AC</i>	VC > AC

Absorption costing vs Variable costing

- **Both U.S. GAAP and IFRS require absorption costing for external reports.**
 - Since absorption costing is required for external reporting, most companies also use it for internal reports rather than incurring the additional cost of maintaining a separate variable cost system for internal reporting.
- Using AC may lead to faulty pricing decisions, keep-or-drop decisions as fixed manufacturing overhead is treated as a variable costs.
- VC is also consistent with CVP analysis.
- Profit under VC is not affected by changes in inventory.
 - AC may potentially produce positive net operating income even when the number of units sold is less than the breakeven point.
 - AC assigns per unit fixed manufacturing overhead costs to production, therefore some fixed manufacturing overhead cost may be deferred to Inventory.

How profit changes with inventory (Actual costing)

Selling price = \$30. No beginning inventory.

Actual costing – Manufacturing OH allocated based on units produced.

Scenario #1 Actual

Produce 25k units, sell 20k units

DM & DL	\$10 per unit
Fixed MOH	\$150k - \$6 per unit

Scenario #2 Actual

Produce 30k units, sell 20k units

DM & DL	\$10 per unit
Fixed MOH	\$150k - \$5 per unit

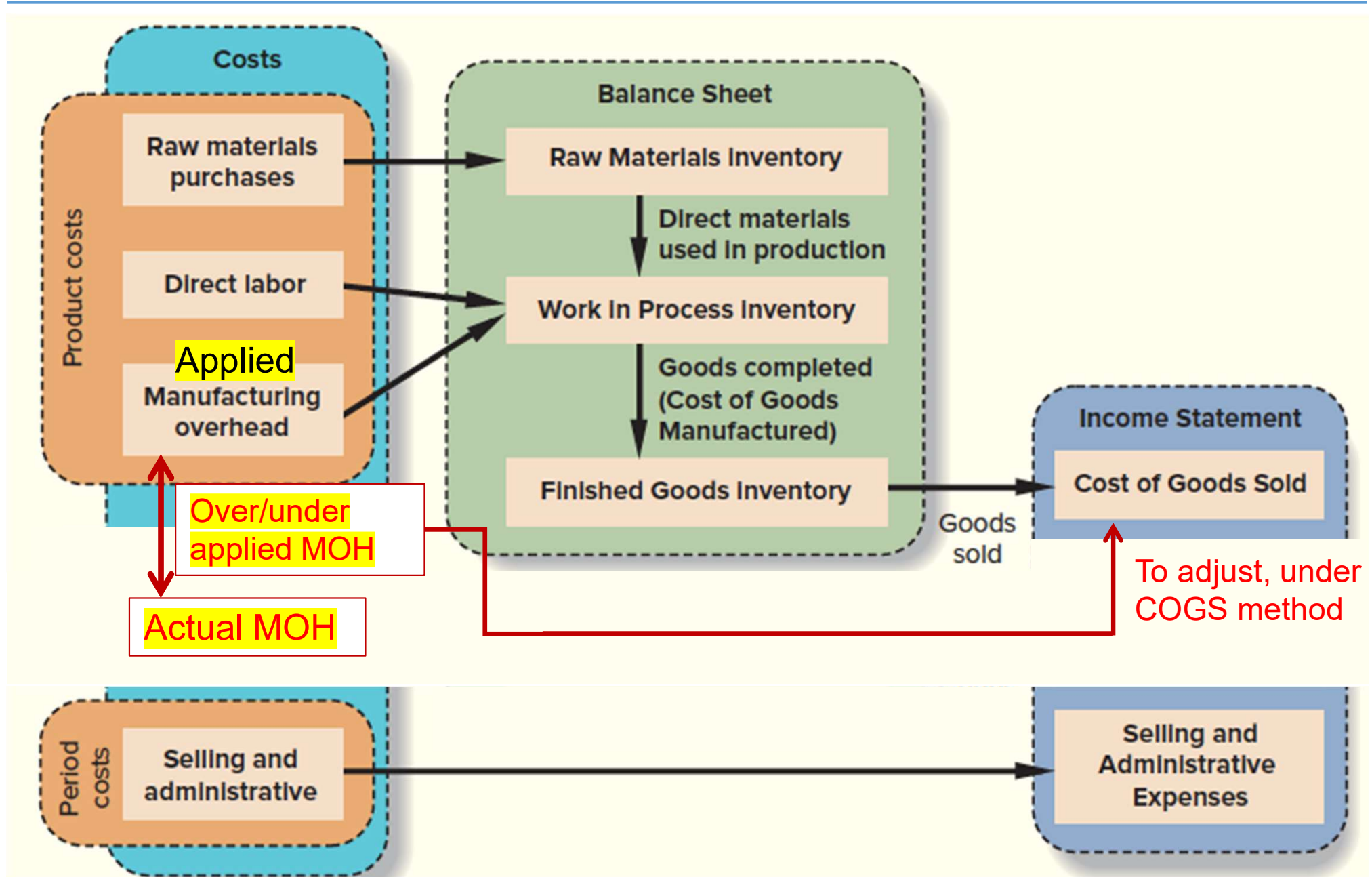
AC	Scenario #1	Scenario #2
Revenue	\$600k	\$600k
COGS	\$320k	\$300k
Gross Profit	\$280k	\$300k
S&A expense	\$160k	\$160k
NOI	\$120k	\$140k

Tweak in change in inventory leads to profit changes.

VC	Scenario #1	Scenario #2
Revenue	\$600k	\$600k
COGS	\$200k	\$200k
Var S&A exp.	\$60k	\$60k
CM	\$340k	\$340k
Fixed MOH	\$150k	\$150k
Fixed S&A exp.	\$100k	\$100k
NOI	\$120k	\$120k

Profit remains unchanged.

Normal Costing



End of topic